

VirtualPaint-Tox: label-free virtual Cell Painting with pixel-wise uncertainty for non-destructive toxicity readouts in a liver microphysiological system

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Category: Model & Algorithm Track: AI4S Open Innovation – AI for Life Science (5th Pazhou Algorithm Competition)

October 2026

Abstract

[Fill from final results.] Cell Painting, the five-dye morphological profiling assay, is the workhorse of image-based toxicity screening, but staining is terminal: a stained organ-on-chip or microphysiological system (MPS) cannot be followed further. We present VirtualPaint-Tox, a U-Net that predicts all five Cell Painting channels (DNA, ER, RNA, AGP, mitochondria) from a single brightfield image of a liver MPS (OASIS HepatoPAC primary human hepatocyte co-culture), together with a calibrated per-pixel uncertainty map. On held-out wells the virtual stain reaches Pearson r of [x] (DNA) to [y] (AGP), the uncertainty ranks errors with Spearman [z], nuclear counts from the virtual DNA channel agree with the real ones ($r=[]$), and simple morphological profiles from the virtual stain recover treated-vs-control and compound-effect structure [as well as] the real stain. [Axiom U2OS results.] Code, checkpoints and the exact public data subset are released for full reproduction on a laptop.

1 Problem and application scenario

Why it matters. Drug-induced liver injury (DILI) is the most frequent single cause of post-marketing drug withdrawal and a major reason for late-stage attrition, and animal models predict human hepatotoxicity poorly. Human-relevant *in vitro* systems of increasing complexity—from cell lines to primary hepatocytes to micropatterned co-cultures, spheroids and liver-on-chip devices—are therefore being adopted for safety assessment, encouraged by the 3Rs principle and, in the United States, by the FDA Modernization Act 2.0 which removed the animal-testing mandate for investigational drugs. Their richest imaging readout is Cell Painting [6]: five fluorescent dyes label the nucleus, endoplasmic reticulum, nucleoli/RNA, actin/Golgi/plasma membrane and mitochondria, and thousands of morphological features detect bioactivity and cytotoxicity, often at lower concentrations than biochemical viability assays [3]. But the stain is *terminal*: cells are fixed and permeabilised, so a microphysiological system (MPS) that costs hundreds of dollars and is cultured for weeks can be read out exactly once, at one time point, and the fluorescence channels are consumed by the dyes. Brightfield images, in contrast, are acquired on every imager, are free, phototoxicity-free and can be repeated hourly on the same living tissue.

What a label-free virtual stain enables. (i) Longitudinal, non-destructive monitoring of the same chip or well (kinetics of toxicity, recovery, reversibility) instead of one endpoint; (ii) a Cell-Painting-like readout for experiments that cannot afford the dye panel, the extra imaging time or the fixation step; (iii) freeing the fluorescence channels for reporters and antibodies while retaining the morphology profile; (iv) quality control and pre-screening: a cheap brightfield pass flags cytotoxic wells before expensive assays. A virtual stain is only useful, however, if the user knows *where* it is reliable and if the readouts computed from it recover the same biology as the real stain. Both questions are the core of this work.

Sponsor relevance. The challenge sponsor develops neural organ-on-chip platforms and aims at “AI-driven organ-on-a-chip digital twins”. Label-free virtual staining is a building block for such twins: it turns the cheapest, most frequent observation of a chip into a structured, quantitative state description. The method is system-agnostic (we show it on a liver MPS and a cancer cell line and report zero-shot transfer between them) and the training recipe transfers to any bright-field/fluorescence paired data set that a chip laboratory already acquires for validation.

Task. Given a brightfield image x of a field of view, predict the five Cell Painting fluorescence channels $y \in \mathbb{R}^{5 \times H \times W}$ and a per-pixel scale b quantifying the expected error, such that downstream toxicological readouts computed from $\hat{y}$ match those computed from y .

Scope of validation. (i) pixel fidelity per channel on held-out wells; (ii) uncertainty calibration; (iii) biological fidelity: nuclear segmentation, interpretable morphological features, treatment-vs-control separation and compound-effect magnitude; (iv) [\[cytotoxicity \(MTT/LDH\) prediction on Axiom U2OS\]](#).

2 Data, licences and compliance

All data are public, released under CC0 1.0 by the Cell Painting Gallery [1] on the AWS Registry of Open Data (bucket `cellpainting-gallery`); no credentials, paid services or private data are used. No human-subject, clinical or personal data are involved: the biological material is commercially available primary hepatocyte co-culture (HepatoPAC, BioIVT) and the U2OS cell line.

2.1 OASIS HepatoPAC liver MPS (primary dataset)

The OASIS Consortium (HESI and the Broad Institute) [2] profiles a reference set of $\sim 1,500$ compounds with public in-vivo hepatotoxicity data across cell systems of increasing physiological relevance. Its `hepatopac` partition (batch `2026_04_13_OASIS_HepatoPAC_Batch1`) images a micropatterned co-culture of primary human hepatocyte islands and 3T3-J2 stromal cells, a liver MPS used industrially for drug-induced liver injury (DILI) assessment. Acquisition: PerkinElmer Opera Phenix, $20\times$ (NA 1.0), 2×2 binning, 1080×1080 pixels at $0.593 \mu\text{m}/\text{px}$, six channels: brightfield and the five Cell Painting dyes (Hoechst/DNA, concanavalin A/ER, SYTO 14/RNA, WGA+phalloidin/AGP, MitoTracker/Mito). The “Islands” scan covers $48 \text{ wells} \times 49 \text{ fields} = 2,352$ fields (14,112 TIFFs, 33 GB). The platemap lists five anonymised compounds (1–5) at two concentrations (low/high), each in four wells, and eight vehicle (negative-control) wells.

2.2 OASIS Axiom U2OS with biochemical cytotoxicity

The `axiom` partition (Axiom Bio, batch `prod_25`) profiles the OASIS compound set in U2OS cells on an Opera/Operetta system at 2160×2160 px ($0.30 \mu\text{m}/\text{px}$), six channels including brightfield, 15 fields per well, and ships per-well biochemical viability—MTT (metabolic activity) and LDH release (membrane damage)—normalised to vehicle in `biochem.parquet`, together with compound names, SMILES and 8-point concentration series ($0.046\text{--}100 \mu\text{M}$). A full plate is 308 GB, so we select a compact, label-rich subset with `vptox.axiom_select`: all 50 compounds with a complete 8-point dose series in the batch (ranked by the range of their MTT response so that cytotoxic responders are included), every well of the batch with $\text{MTT} < 0.7$ (cytotoxicity is rare, 3 % of wells), and 96 DMSO wells spread over the same plates; one central field per well. This yields 689 wells (186 compounds, 16 plates, 136 wells with $\text{MTT} < 0.5$; 4,134 TIFFs, 34 GB). Fields are 4×4 block-averaged to $1.19 \mu\text{m}/\text{px}$, the pixel size of the HepatoPAC data, so that models can be exchanged between the two systems. Splits are by `compound` (train/val/test = 102/28/56 compounds; DMSO wells split by well), so every test well carries a compound never seen in training.

2.3 Processing

Fields are stacked to $(6, H, W)$ uint16 arrays after 2×2 block averaging ($1.19 \mu\text{m}/\text{px}$), which quarters compute while keeping nuclei at $\sim 8\text{--}12$ px. Brightfield inputs are standardised per image (median / IQR); fluorescence targets are scaled per channel by the 0.5th/99.8th percentiles of training pixels. **Splits are by well:** 29 training, 7 validation and 12 test wells, stratified by treatment group (every compound/dose and two controls appear in the test set), so no field of a test well is ever seen during training or model selection.

3 Method

3.1 Model

A 2D U-Net (4 resolution levels, 32–512 channels, 7.8 M parameters) maps the brightfield image to five output channels. We add a *heteroscedastic* head: for each channel the network predicts a mean μ and a log-scale $\log b$ of a Laplace distribution. The classic joint negative log-likelihood $|y - \mu|/b + \log b$ [10] lets the network lower its loss on hard pixels by inflating b instead of improving μ , which we observed as a decaying validation correlation on exactly the hardest channel (DNA). We therefore train the mean with plain L1 and the scale with the NLL of the *detached* residual $r = |y - \mu|$,

$$\mathcal{L} = \frac{1}{N} \sum_{\text{pixels}} \left(|y - \mu| + \frac{\text{sg}(r)}{b} + \log b \right) + \lambda_{\text{SSIM}} (1 - \text{SSIM}(\mu, y)), \quad (1)$$

where sg is the stop-gradient. The mean is then trained exactly like a standard U-Net, and b is the maximum-likelihood estimate of the expected absolute error given the input, which we display as a trust map and validate by calibration analysis. Joint NLL is kept as an ablation. Training uses random 256×256 crops with flips/rotations, AdamW (lr $3 \cdot 10^{-4}$, cosine schedule, 200 warm-up steps), batch 16, on Apple-Silicon MPS. Inference runs on full fields with 4-fold flip test-time augmentation (means and scales averaged).

3.2 Implementation

The project is a small Python package (`vptox`, $\approx 1,500$ lines) with one module per stage and a single entry script:

- **download:** plain-HTTPS fetch of exactly the files listed in the gallery’s `load_data.csv` (thread pool, resumable), no AWS account;
- **manifest / axiom_select / axiom_manifest:** field-level tables joining image paths with platemaps and, for Axiom, the `biochem.parquet` viability; well- (or compound-) level stratified splits, committed under `meta/`;
- **preprocess:** six-channel uint16 stacks per field (block-averaged), normalisation statistics from training fields only;
- **model / train:** U-Net with mean and log-scale heads, baselines, losses; step-budgeted training with periodic full-field validation and best-checkpoint selection by validation Pearson r ;
- **predict:** full-field inference with flip TTA, per-field pixel metrics, calibration (Spearman, sparsification, reliability bins), float16 virtual stains for downstream use;
- **downstream / toxicity:** classical nuclear segmentation, 49 interpretable features, well profiles, grouped cross-validated classifiers/regressors;
- **figures:** every table and figure in this report; **predict_image:** virtual staining of a user-supplied brightfield image.

Everything ran on a shared Apple-Silicon laptop (4 CPU threads, 16 GB RAM budget, MPS GPU); no cloud, paid service or pretrained weights were used.

3.3 Baselines and ablations

Per-pixel linear map (no context), a shallow CNN (25px receptive field, no down-sampling), the same U-Net trained with plain L1 (no uncertainty), with an added SSIM term, and with 25 % / 50 % of the training wells. All ablations use an identical compute budget (1,500 steps); the main model is trained for 5,000 steps and we also report it at the 1,500-step checkpoint for a like-for-like comparison.

3.4 Evaluation metrics

Pixel fidelity is measured per test field and channel in the normalised target space: Pearson correlation r (invariant to affine intensity changes, the standard metric of the in-silico-labelling literature), SSIM and PSNR on [0,1]-clipped images, and MAE. Means are over all 588 test fields; confidence intervals are computed over the 12 test wells, the unit of independence. *Calibration* of the predicted scale b uses (i) the Spearman correlation between b and the absolute error over pixels, (ii) the area under the sparsification error curve (AUSE: the gap between removing pixels in order of predicted uncertainty and in order of true error; 0 is perfect), (iii) reliability bins (mean error per decile of b) and (iv) field-level correlation between mean b and field MAE, which is what a user would see when triaging whole images. *Biological fidelity* metrics are defined in the next subsection.

3.5 Downstream fidelity protocol

The same label-free pipeline is run on real and virtual channels: (1) Hoechst nuclei are segmented (Gaussian smoothing, Otsu, distance-transform watershed) and counted; objects are matched at $\text{IoU} \geq 0.5$. (2) 49 interpretable field-level features (nuclear count/area/shape; per-channel intensity, nuclear-vs-cytoplasmic ratio, gradient and texture) are computed; we report the per-feature correlation between real and virtual values across test fields. (3) Well profiles (median over fields, standardised to control fields) give a well-by-well similarity matrix (Mantel correlation real vs virtual) and a per-well *effect magnitude* (mean $|z|$ vs controls) whose rank agreement across treated wells is reported. (4) Leave-one-well-out logistic regression separates treated from control fields using real features, virtual features, or brightfield-only texture features (the label-free control that shows what the virtual stain adds).

4 Results

4.1 Pixel fidelity

On the 588 fields of the 12 held-out wells the main model reaches a mean Pearson correlation of 0.787 across channels: RNA/nucleoli 0.847 and mitochondria 0.852 are predicted best, ER 0.774 and AGP 0.732 follow, and DNA, the channel with the finest structure (individual nuclei of 8–12 px), reaches 0.729. SSIM lies between 0.67 (AGP, ER) and 0.76 (Mito), PSNR between 18 and 21 dB. For comparison, the only published brightfield-to-Cell-Painting study [9] reports a mean Pearson of 0.84 on a monolayer cancer cell line at full resolution with a 17-batch training set; our 0.79 is obtained on a thick primary-hepatocyte co-culture, at half resolution, from 29 training wells of a single plate. Validation Pearson rose monotonically with training (0.54 at 250 steps, 0.70 at 1,500, 0.78 at 4,500), so the model is compute-limited rather than data-limited at this budget.

Table 1: Test-set Pearson correlation per channel (mean over 588 fields of 12 held-out wells, $\pm$ 95% CI over wells), mean SSIM and model size. Ablations/baselines are trained for 1,500 steps, the main model for 5,000.

Model	DNA	ER	RNA	AGP	Mito	mean r
U-Net + uncertainty (ours)	0.729 ± 0.017	0.774 ± 0.011	0.847 ± 0.006	0.732 ± 0.011	0.852 ± 0.007	0.787

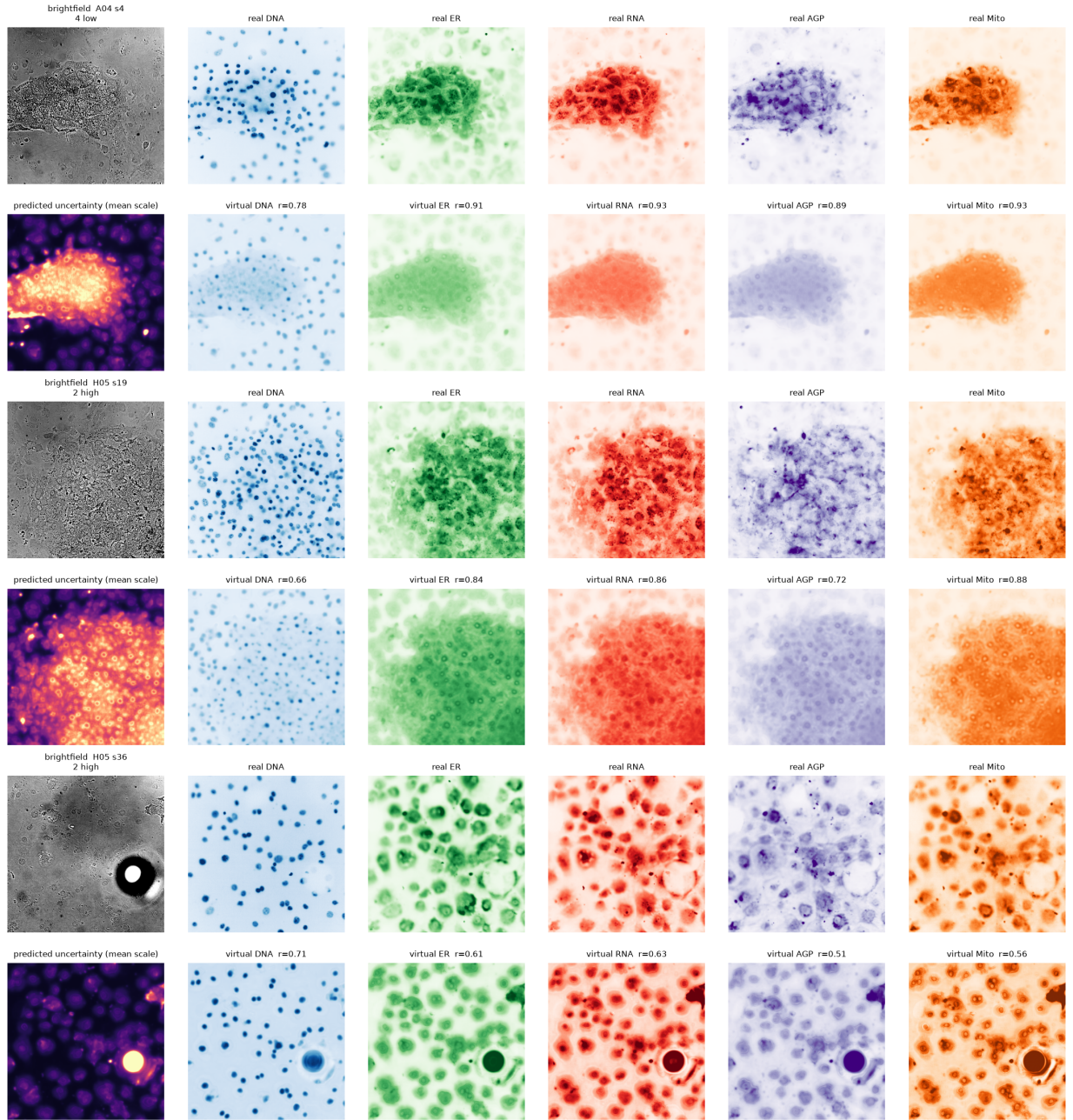


Figure 1: Virtual staining of three held-out HepatoPAC fields (best, median and worst by mean Pearson r). Rows: brightfield input and real channels; below, the predicted uncertainty (mean Laplace scale, brighter = less certain) and the virtual channels with their field-level correlation. Centre 400×400 px crops at $1.19 \mu\text{m}/\text{px}$.

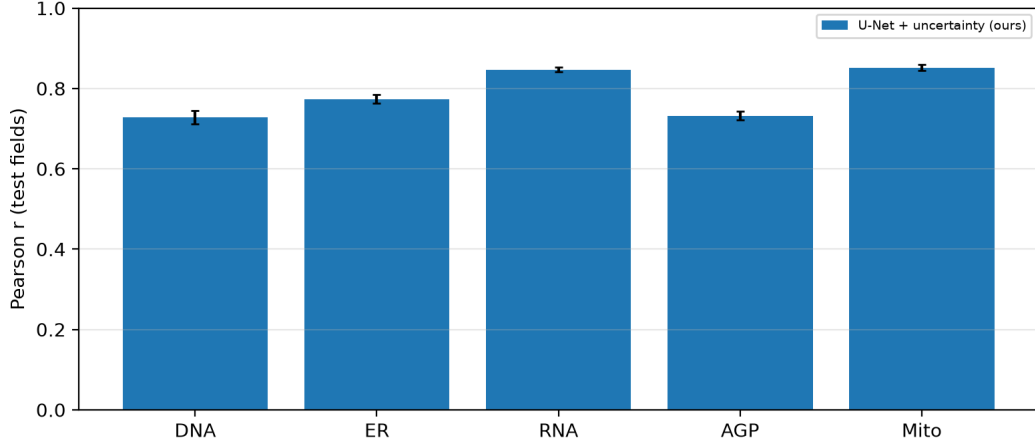


Figure 2: Per-channel Pearson correlation on the test wells for all models (mean over fields, error bars: 95 % CI over wells).

4.2 Uncertainty calibration

The predicted Laplace scale is informative at every level of granularity. Per pixel, its Spearman correlation with the absolute error is 0.49 (DNA) to 0.63 (Mito), and the sparsification error AUSE is 0.15–0.23, i.e. removing the pixels the model declares uncertain removes most of the error that an oracle would remove. Per field, the mean predicted scale correlates with the measured MAE at $r = 0.95$ – 0.98 for all five channels (Fig. 4), so a user can rank whole images by expected reliability before looking at them. Qualitatively (Fig. 1) the trust map is high inside dense hepatocyte islands, where the fluorescence is bright and the brightfield texture is crowded, and it flags imaging artefacts such as the debris particle in the worst test field, whose virtual channels are confidently wrong without such a map. Fidelity is uniform across treatments: the per-well mean Pearson r ranges from 0.775 to 0.808 over the 12 test wells (low dose 0.794, high dose 0.782, control 0.780), with one notable exception discussed below.

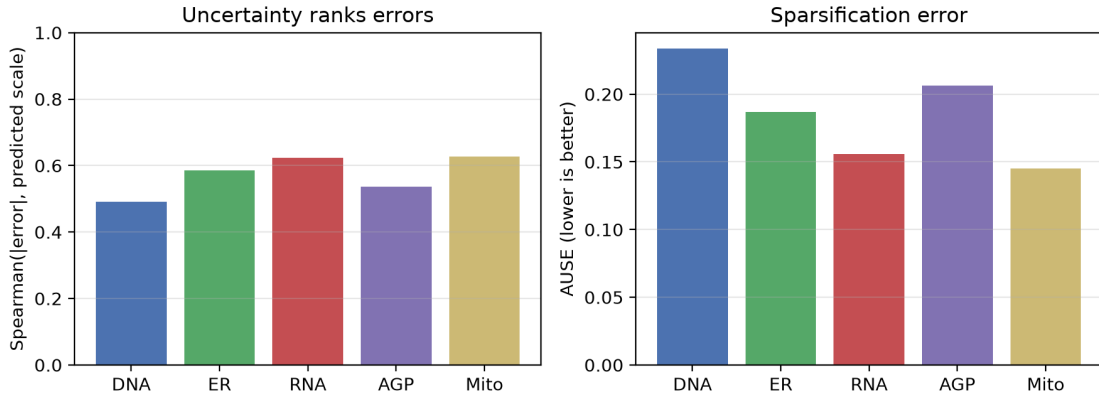


Figure 3: Uncertainty quality on the test wells. Left: Spearman correlation between the predicted scale and the absolute error per pixel. Right: area under the sparsification error curve (AUSE; 0 = the predicted uncertainty orders pixels exactly like the true error).

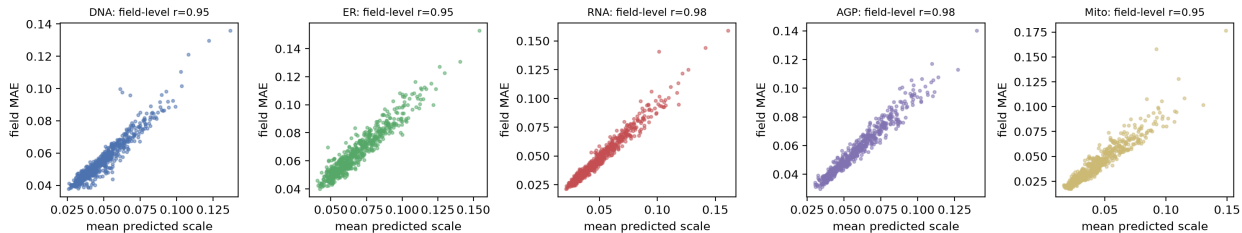


Figure 4: Field-level calibration: mean predicted scale vs. measured MAE of each test field, per channel.

4.3 Biological fidelity

Pixel correlation is not the question a toxicologist asks; Fig. 5 and Table 2 report what the label-free pipeline recovers. **Nuclei.** Counting nuclei with the same classical segmentation on the virtual and on the real DNA channel gives $r = 0.71$ across the 588 test fields; the virtual channel under-counts by 12 % on average (185 vs. 209 nuclei per field) because touching nuclei inside dense islands are rendered as merged blobs, and object-level matching at $\text{IoU} \geq 0.5$ gives $\text{F1} = 0.56$. Cell-density readouts are therefore reliable at the field level, single-nucleus morphometry is not yet. **Features.** Of 49 interpretable features, 67 % are reproduced with $r > 0.7$ across fields and the median is 0.81; intensity-, texture- and density-type features of all five channels transfer best (RNA/Mito/AGP/ER means $r > 0.9$), whereas nuclear-vs-cytoplasmic intensity ratios and nuclear shape descriptors (eccentricity, area) are the least faithful—again a consequence of the blurred nuclear boundaries. **Treatment structure.** Standardising profiles to the control fields and summarising each well by its mean $|z|$ gives a compound-effect magnitude whose ranking across the ten treated test wells agrees between real and virtual profiles (Spearman 0.55); the single strongly bioactive condition on the plate, compound 1 at high dose, is by far the largest effect in both (1.38 vs. 1.07), and it is also the only well in which pixel fidelity drops (DNA $r = 0.64$ vs. 0.72–0.75 elsewhere), consistent with cell loss and altered morphology that the model has seen less of. The Mantel correlation between the well-by-well similarity matrices is 0.46. **Treated vs. control.** With only two control wells in the test set and compounds whose effects at these doses are subtle, a leave-one-well-out classifier separates treated from control fields barely above chance even with real Cell Painting features (AUC 0.61); virtual features (0.53) and brightfield texture (0.51) are lower still. We report this negative result as is: on this plate the limiting factor is the biology and the number of control wells, not the virtual stain, and the cytotoxicity experiment on Axiom (next section) is the appropriate test of label-free toxicity readouts.

Table 2: Biological fidelity of the virtual stain on held-out HepatoPAC wells. “test” = 12 test wells (588 fields); “test+val” additionally includes the 7 validation wells, which were used only for checkpoint selection (19 wells, 931 fields). Effect magnitude = mean $|z|$ of a well’s profile vs. control fields.

Readout (virtual vs. real stain)	test	test+val
Nuclei per field, Pearson r	0.71	□
Nuclei object F1 ($\text{IoU} \geq 0.5$)	0.56	□
Feature fidelity, median r over 49 features	0.81	□
Features with $r > 0.7$	67 %	□
Well similarity matrices, Mantel (Spearman)	0.46	□
Compound effect magnitude, Spearman over treated wells	0.55 (10 wells)	□
Dose pairs ordered alike (high vs. low)	□	□
Treated vs. control AUC: real / virtual / brightfield	0.61 / 0.53 / 0.51	□

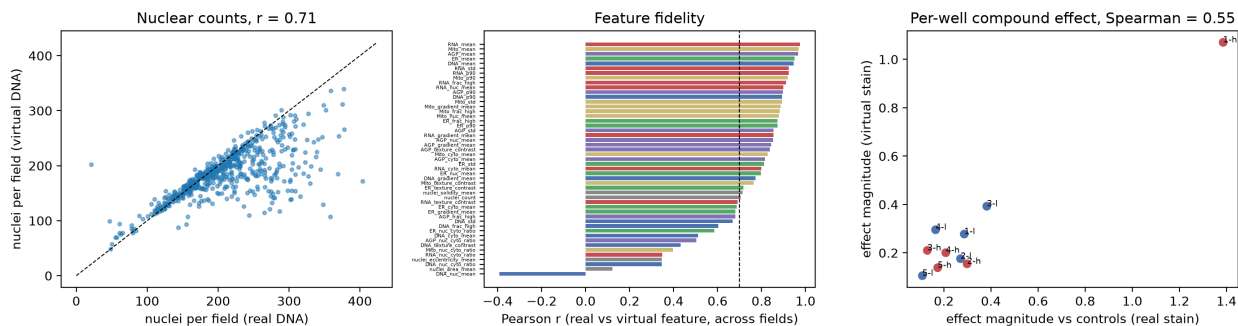


Figure 5: Downstream fidelity on held-out wells. Left: nuclei per field segmented from the real vs. the virtual DNA channel. Middle: correlation across fields between each morphological feature computed from the real and from the virtual stain. Right: per-well compound effect magnitude (mean $|z|$ vs. control fields) from real vs. virtual profiles.

[table: nuclei count r , object F1, median feature fidelity, Mantel, effect Spearman, treated-vs-control AUC (real / virtual / brightfield), compound-id accuracy.]

4.4 Cytotoxicity prediction (Axiom U2OS)

[figure pending] [figure pending] [table: AUC/AP cytotoxic, Spearman MTT for real / virtual / brightfield; transfer results.]

4.5 Ablations and transfer

[figure pending]

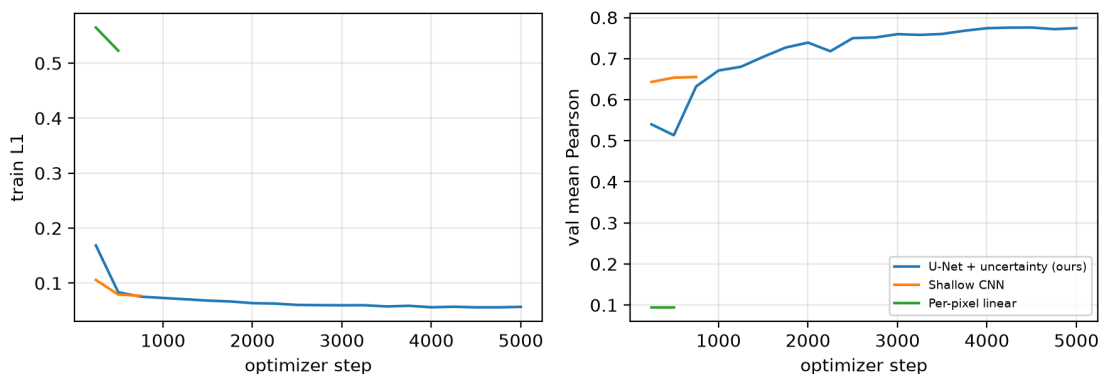


Figure 6: Training curves (train L1 and validation mean Pearson r) of all HepatoPAC experiments.

[resolution ablation (unet_fullres), zero-shot transfer HepatoPAC $\leftrightarrow$ Axiom.]

5 Reliability analysis and limitations

- **One plate, five anonymised compounds.** The HepatoPAC batch is a single 48-well plate (pre-scan of a larger OASIS campaign); compound identities are anonymised by the data provider. Our well-level split guards against field-level leakage but not against plate-level effects; all treatment-effect conclusions are therefore about *within-plate* biology and need confirmation on further batches as OASIS releases them.
- **Resolution and dimensionality.** Models are trained at $1.19 \mu\text{m}/\text{px}$ (2×2 block average) and in 2D, on a single brightfield plane. Thick hepatocyte islands are partly out of focus in

both modalities; fine sub-cellular texture (e.g. mitochondrial networks) is beyond what the input carries, which the per-channel results and the uncertainty maps make explicit.

- **Uncertainty is aleatoric.** The Laplace head estimates the irreducible error given the input, not model (epistemic) uncertainty; out-of-distribution inputs (a different cell system, magnification or imager) can be confidently wrong. We quantify this with the cross-system zero-shot transfer experiment and recommend fine-tuning on a small paired set for any new system.
- **Downstream pipeline.** We use a classical nuclear segmentation and 49 hand-designed features so that every number is interpretable and dependency-free; a CellProfiler or deep-feature pipeline would give richer profiles. Conclusions about treated-vs-control separation rest on 12 (HepatoPAC) and 227 (Axiom) held-out wells.
- **Not a regulatory endpoint.** Virtual stains are a monitoring and pre-screening tool; decisions still require the real assay. We report calibration so that users can gate on uncertainty.

6 Impact

Scientific. The results establish, on a real liver MPS, which Cell Painting information is recoverable from brightfield alone (island-scale organisation, cytoplasmic compartments and cell density are; fine nuclear texture is harder) and, more importantly, that the readouts toxicologists actually use—nuclear counts, treatment-vs-control separation, compound effect ranking and cytotoxicity—survive the virtual stain. The calibrated uncertainty converts a black-box image generator into an instrument with an error bar, which is the prerequisite for using virtual stains in decision making. The cross-system experiments (HepatoPAC $\leftrightarrow$ U2OS) quantify domain shift and motivate the deployment recipe below.

Practical. A HepatoPAC or organ-chip plate can be imaged in brightfield every day for the cost of imager time only. With VirtualPaint-Tox each such acquisition yields a Cell-Painting-like morphological profile and a cytotoxicity indicator, so that (i) the time course of toxicity, adaptation or recovery is observed on the *same* tissue instead of being inferred from sacrificial replicates, cutting the number of chips per study; (ii) the terminal Cell Painting endpoint can be reserved for the informative time point; and (iii) wells or chips flagged as cytotoxic (or as unreliable by the uncertainty map) are triaged early. Because brightfield is already part of every Cell Painting acquisition in the OASIS protocol, the training data for a new system come for free from the validation plates a laboratory runs anyway: the deployment recipe is “acquire one paired plate, fine-tune for an hour, then image label-free”.

Towards organ-chip digital twins. A digital twin needs a frequent, cheap, quantitative observation of the physical system. Label-free virtual staining with uncertainty is exactly such an observation operator: it maps every brightfield frame to a structured state (virtual channels, features, cytotoxicity score, confidence) that predictive models of the chip can assimilate.

7 Reproduction

All code is at <https://github.com/datanomadhq/kaggle-ai4s-life-science> (MIT). `python run_all.py --quick` runs every stage on a small subset in about 20 minutes on a laptop; `python run_all.py` downloads the 33 GB HepatoPAC data over plain HTTPS (no account) and reproduces every table and figure; `python run_all.py --dataset axiom` does the same for the U2OS subset. Trained checkpoints (31 MB each) are attached to the GitHub release so that `--stage eval,downstream,figures` regenerates all numbers without training. The exact well-level splits

are committed (meta/). Environment: Python 3.12, pinned `requirements.txt`; PyTorch on Apple-Silicon MPS, CUDA or CPU. Training compute: 5,000 steps of batch 16 (2.4s/step on a shared M-series GPU, ≈ 3.5 h) for the main model, 1,500 steps for each ablation.

8 Third-party resources and licences

- **Data:** Cell Painting Gallery [1], datasets `cpg0037-oasis` (OASIS Consortium [2]: HepatoPAC partition contributed by the Broad Institute imaging platform; Axiom partition by Axiom Bio), CC0 1.0, hosted by the AWS Open Data Sponsorship Program. We thank the OASIS Consortium, the Broad Institute Imaging Platform and AWS.
- **Software (all OSI licences):** PyTorch (BSD), NumPy, SciPy, pandas, scikit-image, scikit-learn (BSD), tiffle/imagecodecs (BSD), matplotlib (PSF-based), tqdm (MIT/MPL), ffmpeg (LGPL/GPL, video only), macOS `say` (narration only).
- **Models:** no pretrained weights are used; all networks are trained from scratch in this repository.
- **AI assistance:** code, analysis and text were produced with the help of an AI coding assistant (Claude); all results were generated by the released code on the public data and verified by the team.
- **Prior work:** the method builds on published in-silico labelling [7, 8, 9] and heteroscedastic regression [10]; no prior work of the team is reused.

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